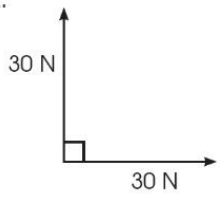
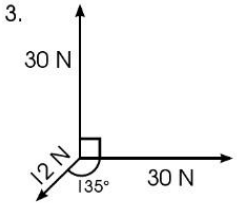
Chapter1 Turing effect of force

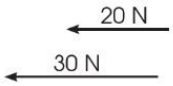
I. Adding parallel forces and Non-Parallel Forces

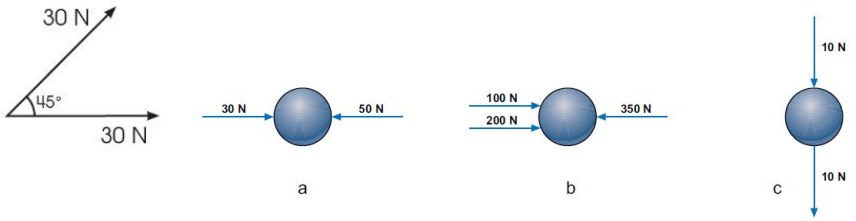
1. Find the resultant force in each diagram and draw the resultant vector. Use a ruler and a protractor where necessary. Scale: 1 cm = 10 N, where N represents newtons of force.

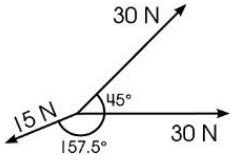
1. 

2. 

3. 

4. 

5. 

6. 

2. You walk 3.0 km due north, and then 4.0 km due east.
a Calculate the total distance in km you have travelled.
b Make a scale drawing of your walk, and use it to find your final displacement. Remember to give both the magnitude and the direction.

Given: $\tan^{-1}\left(\frac{4}{3}\right) \approx 53.1^{\circ}$, $\tan^{-1}\left(\frac{3}{4}\right) \approx 36.9^{\circ}$

3. You walk 5.0 km due west and then 12.0 km due south.
a) Calculate the total distance in km you have traveled.
b) Make a scale drawing of your walk, and use it to find your final displacement, including both magnitude and direction.

Given: $\tan^{-1}\left(\frac{12}{5}\right) \approx 67.4^{\circ}$, $\tan^{-1}\left(\frac{5}{12}\right) \approx 22.7^{\circ}$

4. You walk 8.0 km due east and then 6.0 km due north.
a) Calculate the total distance in km you have traveled.
b) Make a scale drawing of your walk, and use it to find your final displacement, including both magnitude and direction.

Given: $\tan^{-1}\left(\frac{6}{8}\right) \approx 36.9^{\circ}$, $\tan^{-1}\left(\frac{8}{6}\right) \approx 53.1^{\circ}$

5. A student walks 8.0 km south-east and then 12 km due west.
a Draw a vector diagram showing the route.
b Use your diagram to calculate the resultant displacement and direction of resultant displacement.

Given: $\tan^{-1}\left(\frac{12}{8}\right) \approx 56.3^{\circ}$, $\tan^{-1}\left(\frac{8}{12}\right) \approx 33.7^{\circ}$, $\tan^{-1}\left(\frac{5.66}{6.34}\right) \approx 42.4^{\circ}$

6. A hiker's displacement is 15 km [40° E of N]. What is the north component of his displacement?
Ans: 11 km [N]

7. A snowmobile travels 65 km [37° E of S]. How far east does it travel? **Ans: 39 km [E]**

Chapter1 Resolving force & Turing effect of force

II. Resolving force

1. A brick with a weight of 20.0 N is on a slope at 25° to the horizontal.
a) Find the size of the force acting on it down the slope.
b) Find the size of the reaction force exerted on it by the slope.
Given: $\cos 25^\circ = 0.91, \sin 25^\circ = 0.42$

2. A 30.0 N box is resting on a ramp inclined at 35° to the horizontal.
(a) Find the size of the force acting on the box down the ramp.
(b) Find the size of the reaction force exerted by the ramp on the box.
Given: $\cos 35^\circ = 0.82, \sin 35^\circ = 0.57$

3. A 50.0 N sled is placed on a hill inclined at 40° to the horizontal.
(a) Find the size of the force acting on the sled down the hill.
(b) Find the size of the reaction force exerted by the hill on the sled.
Given: $\cos 40^\circ = 0.77, \sin 40^\circ = 0.64$

4. Jim shovels snow after a storm by exerting a force of 60N on his shovel at an angle of 30^0 to the vertical axis. What are the horizontal and vertical components of the force exerted by Jim?
Given: $\cos 30^\circ = \frac{\sqrt{3}}{2}, \sin 30^\circ = \frac{1}{2}$

5. Maria pushes a box with a force of 100 N at an angle of 40° to the vertical.
(a) Find the horizontal component of the force exerted by Maria.
(b) Find the vertical component of the force exerted by Maria.
Given: $\cos 40^\circ = 0.77, \sin 40^\circ = 0.64$

6. John pulls a cart by applying a force of 80 N at an angle of 25° to the vertical.
(a) Find the horizontal component of the force exerted by John.
(b) Find the vertical component of the force exerted by John.
Given: $\cos 25^\circ = 0.91, \sin 25^\circ = 0.42$

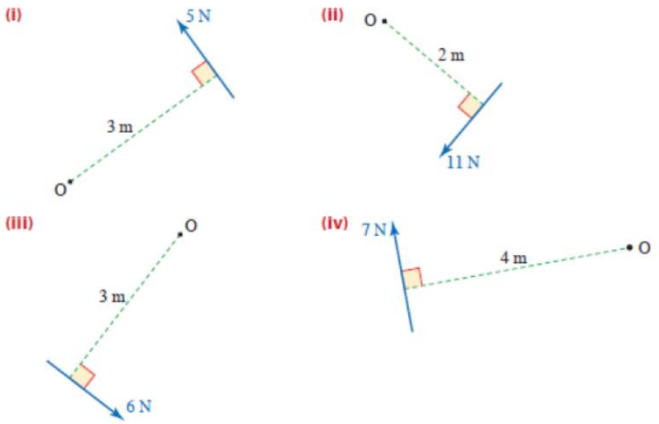
7. To open a door, a force of 15 N is applied at an angle of 30° to the horizontal. Determine the horizontal and vertical components of the force.

III. Turning effect of a force

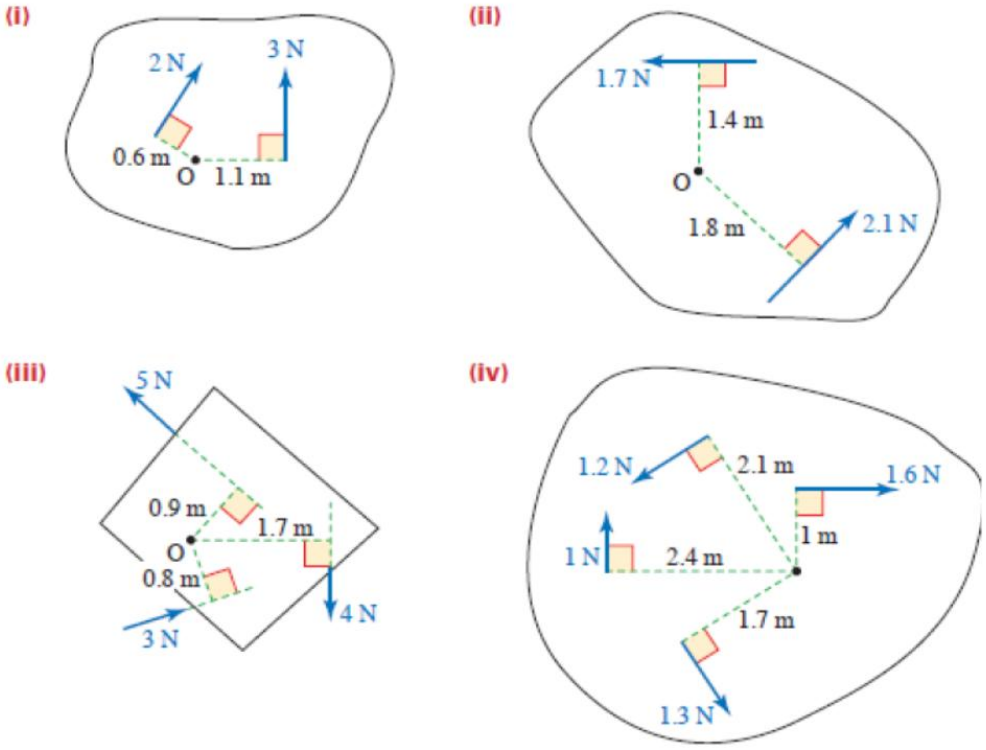
8. Find the moment provided by a force of 73.1 N acting at a perpendicular distance of 0.25 m from a pivot.

9. An airtight chamber is kept shut by a release wheel with a diameter of 0.35 m. The release wheel won't open unless a moment of at least 50.0 Nm is applied across it. What's the minimum force needed to open the chamber door?

10. In each of the situations shown below, find the moment of the force about the point and state whether it is positive (anticlockwise) or negative(clockwise).



11. The situations below involve several forces acting on each object. For each one, find the total moment.



Chapter1 Turing effect of force & Principle of moment

IV. Turning effect of a force

1. The situations below involve several forces acting on each object. Determine the total moment about fixed point O and state whether it is clockwise or anticlockwise for each diagram as following.

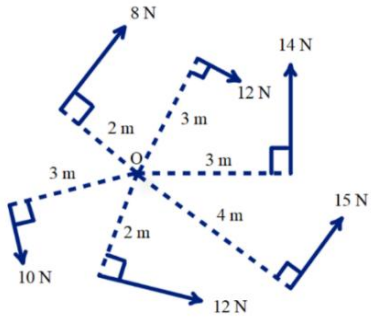


Figure A

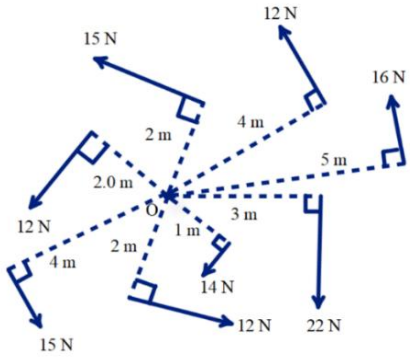
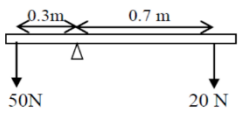
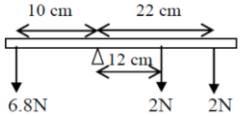


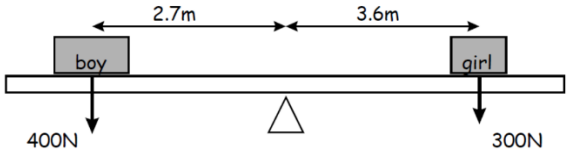
Figure B

2. Calculate the total clockwise moment and the total anticlockwise moment in the following cases. Hence, determine if the rod will balance, turn clockwise or anticlockwise.

Diagram	Total clockwise moment	Total Anti-clockwise moment	Net Moment
			
			

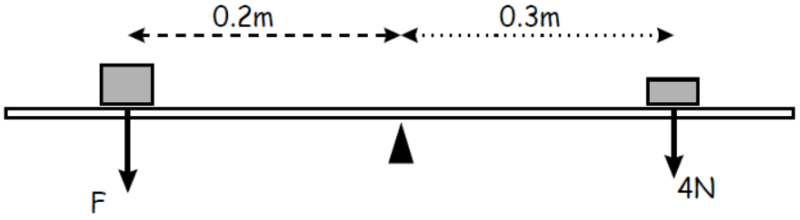
V. Principle of Moment

1. The diagram represents a boy of weight 400N sitting on the left of a seesaw, 2.7m away from the pivot.

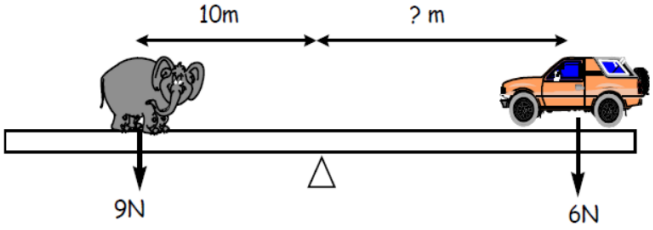


1) Who is trying to turn the seesaw clockwise? _____
2) Who is trying to turn the seesaw anticlockwise? _____
3) What is the clockwise moment? _____
4) What is the anticlockwise moment? _____
5) Is the seesaw balanced? _____

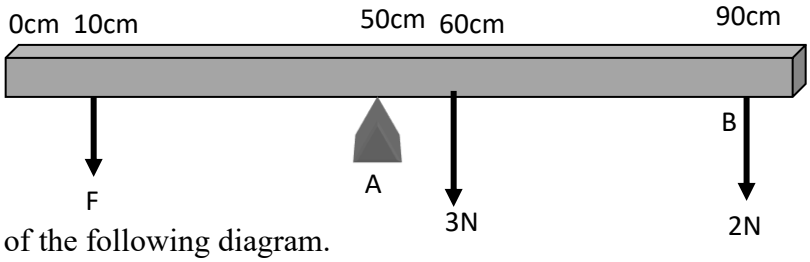
2. The diagram below shows a ruler in equilibrium, What is the value of the force F which is balancing the 4N force?



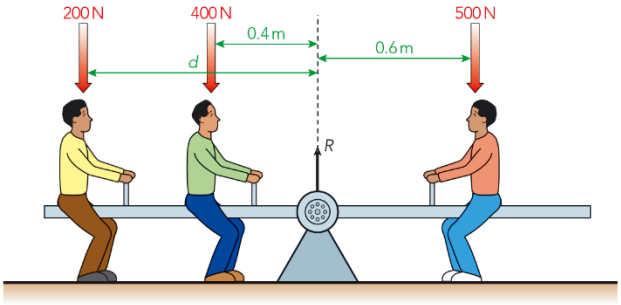
3. The diagram below shows two toys being balanced by a young girl, find the distance of the truck from the pivot for the system to be balanced.



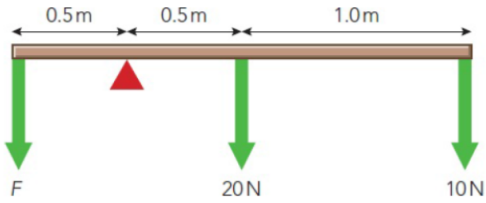
4. A uniform beam of 1m is being balanced as shown below. Determine the unknow force F to ensure rotation equilibrium.



8. Calculate the distance *d* of the following diagram.



9. The beam shown in Figure is 2.0 metres long and has a weight of 20 N. It is pivoted as shown. A force of 10 N acts downwards at one end. What force, *F* must be applied downwards at the other end to balance the beam?



10. A uniform metre rule is pivoted at its centre P, and 3 masses are placed at A, B and C as shown in Fig. 3.10. Find the value for the weight W of the mass M placed at C so that the metre rule is balanced horizontally.

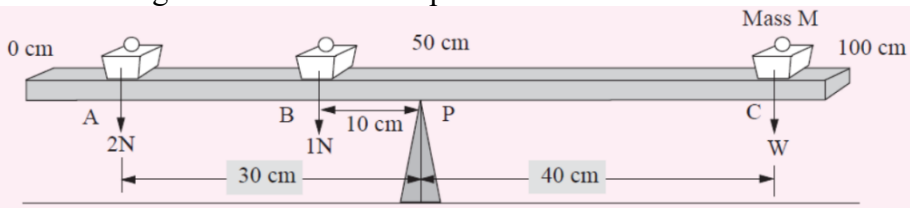


Fig. 3.10: Principle of moment

VI. Parallel Forces

11. A 400 N child and a 300 N child sit on either end of a 2m long seesaw. Where along the seesaw should the pivot support be placed to ensure rotational equilibrium?

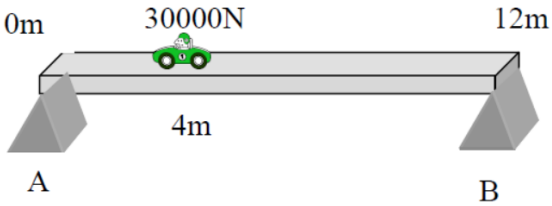
12. A uniform meter stick, supported at the 30cm mark, is balanced when a 0.50 N weight is hung at the 0.0 cm mark. What is the weight of the meter stick?A pivot at the 35cm mark supports a meter stick. It is brought into balance by a 0.85 N weight suspended at the 5.5-cm mark. Compute the weight of the meter stick.

13. A uniform meter stick weighs 1.4N, and is supported by a pivot at the 65cm mark. Where must the 0.85 N force be suspended to balance the stick?

14. A 50cm uniform meter stick is suspended by two loads 600N and 400N at both ends. Find the resultant force and location of resultant force.

15. A 100cm meter rule is exerted by three forces with the same directions. The right end is exerted by 20N, the left end is exerted by 90N and at the middle is exerted by 30N. Find the net force.

16. A 12m long uniform bridge of negligible weights is being supported at the 2 ends. A car weighing 30000N parked at the 4m mark as shown. Replace the weight of car by two parallel force at A and B.



17. A 10m light beam is supported at the ends A and B. The weight of 40000N is placed at 4m from A. Find the downward forces exert on the supports A and B.